1. Aim

To write a Java program that checks whether the given brackets are balanced. The program supports round brackets **()**, curly brackets **{}**, and square brackets **[]**.

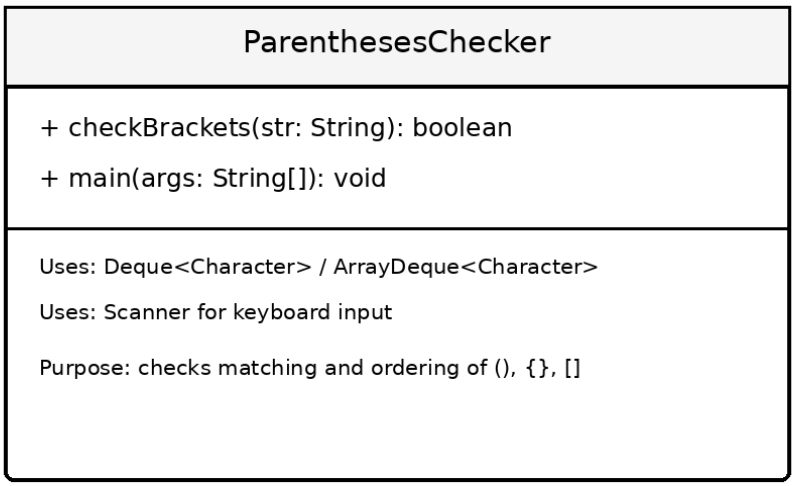
2. Description

The program reads a string from the user and checks each character. Opening brackets are pushed onto a stack. When a closing bracket is found, the program checks whether the stack is empty and whether the most recent opening bracket matches it. If a mismatch is found, the method returns **false**. At the end, the brackets are balanced only if the stack is empty.

3. Software Requirements

Requirement	**Details**
Hardware	Computer or laptop
Operating system	Windows, Linux, or macOS
Software	JDK 17 or later; any Java IDE or text editor
Input	A string containing brackets
Output	Balanced or not-balanced message

4. UML Diagram



5. Source Code

File name: **ParenthesesChecker.java**

```
package recordprograms;

import java.util.ArrayDeque;
import java.util.Deque;
import java.util.Scanner;

public class ParenthesesChecker
{
    public static boolean checkBrackets(String str)
    {
        Deque<Character> stack = new ArrayDeque<>();

        for (int i = 0; i < str.length(); i++)
        {
            char ch = str.charAt(i);

            if (ch == '(' || ch == '{' || ch == '[')
            {
                stack.push(ch);
            }
            else if (ch == ')' || ch == '}' || ch == ']')
            {
                if (stack.isEmpty())
                    return false;

                char open = stack.pop();

                if (ch == ')' && open != '(')
                    return false;

                if (ch == '}' && open != '{')
                    return false;

                if (ch == ']' && open != '[')
                    return false;
            }
            else
            {
                return false;
            }
        }

        return stack.isEmpty();
    }

    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);

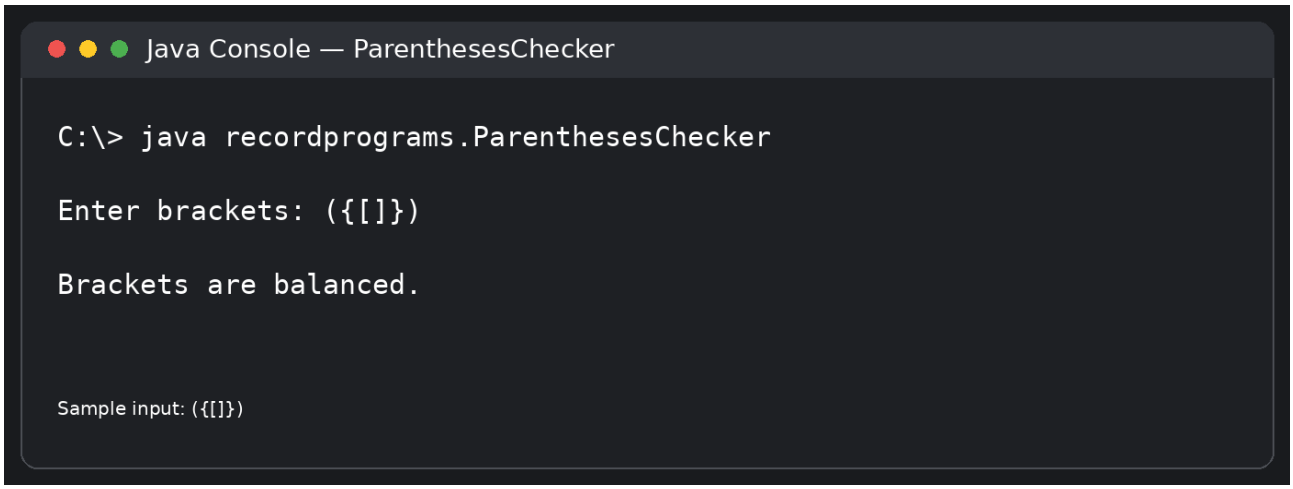
        System.out.print("Enter brackets: ");
        String brackets = sc.nextLine();

        if (checkBrackets(brackets))
            System.out.println("Brackets are balanced.");
        else
            System.out.println("Brackets are not balanced.");

        sc.close();
    }
}
```

6. Output Screenshot

Sample input: **({[]})**



The sample input contains correctly matched and properly nested brackets, so the program displays “Brackets are balanced.”

7. Result

Thus, the Java program to check balanced parentheses using a stack was written and tested successfully with the sample input.

Note: The code rejects any non-bracket character because of the final else block. Enter only (), {}, and [] characters as input.